

Milestone 1 Progress Evaluation

AI Task Agent Using More Flexible Generic Tools via Model Context Protocol (MCP)

1. Project title, names and email addresses of team members (CSE members first)

AI Task Agent Using More Flexible Generic Tools via Model Context Protocol (MCP)

Team Members

Jadon Duff, Computer Science (jduff2024@my.fit.edu)

Jacob Ewasko, Computer Science (jewasko2024@my.fit.edu)

Ioannis Limniatis, Computer Science (ilimniatis2023@my.fit.edu)

Jacob Lebkuether, Software Engineering (jlebkuether2023@my.fit.edu)

2. Faculty Advisor: name and email address

Eraldo Ribeiro, College of Engineering and Science: Department of Electrical Engineering and Computer Science (eribeiro@fit.edu)

3. Client (or user): name and affiliation (department at FIT, company name, ...)

Doug Armstrong, District Directorate Chief - Response of The United States Coast Guard Auxiliary

4. Progress of current Milestone (progress matrix)

Task	Completion %	Jadon	Jacob E	Jacob L	Ioannis	To Do
1. Compare and Select Technical Tools	100%	25%	25%	25%	25%	None
2. "Hello World" Demos	100%	25%	25%	25%	25%	None
3. Resolve Technical Challenges	100%	25%	25%	25%	25%	None
4. Compare and Select Collaboration Tools	100%	25%	25%	25%	25%	None
5. Requirement Document	100%	50%	20%	10%	20%	None
6. Design Document	100%	20%	10%	50%	20%	None
7. Test Plan	100%	15%	50%	15%	20%	None

5. Discussion (at least a paragraph) of each accomplished task (and obstacles) for the current Milestone:
- Task 1 (Compare and Select Technical Tools): For the compare and select technical tools, the team divided the projects into sections that would require technical tools. For each of these sections/goals/features, the team identified what tool format would accomplish the goal. The team met in person to go over the technical tools, with each group member creating several comparison tables for their assigned tools, and the group voted on a tool for each category based on the table's information. Some obstacles the group encountered were for topics they were unfamiliar with, such as Ubuntu and MCP, and required further research to determine a good tool.
 - Task 2 ("Hello World" Demos): The group then did quick demos for their assigned tools after they were compared and selected. The group did some demos of new tools in-person (to solve this obstacle since they were new), such as the local LLM, and did the remaining demos virtually. The team noted their findings to the wider group, which allowed each group member to gain a thorough understanding of each tool and what it can do.
 - Task 3 (Resolve Technical Challenges): To resolve technical challenges, the team members either noted their answers to the technical challenges after research in the team's Confluence space, or the team discussed their technical challenges as a whole group, and how to solve them. Technical challenges were assigned to the team member who knew the topic best, which helped address obstacles since challenges were assigned to their corresponding experts. Results were shared with the full team to ensure all team members were on the same page.
 - Task 4 (Compare and Select Collaboration Tools): The team completed this task similar to the compare and select technical tools task. The team members created a table highlighting the pros and cons of various collaboration tools. Then, the team voted on the desired collaboration tools and set them up.
 - Task 5 (Requirements Document): For the requirements document, Jadon was assigned as the primary "owner." This means that he started the document & completed the draft. The team was relatively new to standardized, official software requirements, which was an obstacle in completing this task. Requirements were split into Functional Requirements, Interface Requirements, and Performance Requirements. Each was worked through to identify the requirements the system needed to meet, using the features and project plan as a baseline.
 - Task 6 (Design Document): For the requirements document, Jacob L was assigned as the primary "owner." This means that he started the document by creating the base formatting and introduction. Team members each wrote several sections of the document and revised the content and graphics. Jacob L wrote the introduction, requirements matrix, user interface design, and data design sections. Iamantis wrote the system overview, architectural design, decomposition description, design rationale, and component design. Jadon created both architecture diagrams. Jacob L created the GUI mockup.
 - Task 7 (Test Plan): For the test plan document, Jacob E was assigned as the primary "owner". This means that he started the document and completed the

draft. Jadon gave feedback on how the Test Plan document should look like, including tests that map to multiple requirements.

6. Discussion (at least a paragraph) of contribution of each team member to the current Milestone:

- Jadon Duff: Contributions include 3 AI-related technical tool evaluations (LLM, RAG, Deployment Environment) and a Collaboration tool (Task Calendar). Four hello world demos of the prior tools were demoed, including an in-person demo of a local LLM to the team. Resolved 10 of the team's technical challenges while assisting other teammates with their technical challenges, providing a write-up for each of the technical challenges. Discussed the write-up with the team and added to team's internal documents repository. Served as the document owner for the requirements, mapping 130+ functional, interface, and performance requirements for the team. Designed the system architecture diagram, high level component diagram, and overall system diagram. Assisted team in mapping requirements to tests. Attended weekly meetings with the advisor alongside the team.
- Jacob E: Contributions include the decisions of document and presentation collaboration tools, decisions of the Browser, Email, Agentic Dynamic Memory, and GUI technical tools, "Hello World" Demos for the Browser, Email, Agentic Dynamic Memory, and GUI, resolving some technical challenges, and a majority of the Test Plan document, where the requirements are planned to be tested in the upcoming future. Further contributions include finishing up the Requirements document as well.
- Jacob L: Contributions include the "Hello World" Demos for the Linux, Communications, and Fast Context. Compare and select collaboration tools for software development. Compare and select technical tools for Linux, communications, and fast context. Served as the design document owner. Wrote base structure of the software design document, table of contents, requirements matrix, introduction, data design, and user interface design sections. Reviewed requirements alongside the team. Reviewed the project's test plan alongside the team. Helped in managing the team's Jira by creating, editing, and managing work items. Helped in managing the team's Confluence by organizing documents and editing the knowledge base when necessary. Attended weekly meetings with the advisor alongside the team.
- Ioannis: Contributions include the Hello World demos for the MCP servers, PDF reading, and Context Management windows. Compare and gather available tools for MCP servers, PDF Reading, and Context Window Management. Organized and help write comparisons for group communication tools. Helped writing several sections mainly the constraints for the requirements document, as well as bulking up and reviewing the requirements. Discussed and assisted with several paragraphs throughout. Communicated with the team as well as helped work through the writing and paragraphs of the design document.

7. Plan for the next Milestone (task matrix): Milestone 2

Task	Jadon	Jacob E	Jacob L	Ioannis
1. Implement, test, and demo Prototype for Task Execution using Generic MCP Tools, without advanced security features implemented (feature 1).	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.	Implement/test/demo Model Context Protocol tools to meet at minimum 20% of the ATA-FTE requirements.
2. Implement, test, and demo Prototype AI Agent, with full connection to Prototype Task Execution using Generic MCP Tools (feature 1).	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.	Implement/test/demo LLM connected to AI Agent to solve at least 3 of the ATA-OPS requirements.
3. Implement, test, and demo POC RAG memory, with basic documentation tested and linked to Prototype AI Agent (feature 2).	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.	Implement/test/demo Proof-of-Concept RAG system that allows AI Agent to ingest documents and media.
4. Implement, test, and demo Prototype for Versatile Deployment with Widespread Hardware	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to	Implement/test/demo prototype for deployment. Meaning, implement/test/demo a user interface, allowing users to

Compatibility, with the system functional to current development on a new install on a new system (feature 3).	interact with the system.	interact with the system.	interact with the system.	interact with the system.
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8. Discussion (at least a paragraph) of each planned task for the next Milestone or Discussion of at least 5 "Lessons Learned" if this is for Milestone 6
 - Task 1: The plan for “Implement, test, and demo Prototype for Task Execution using Generic MCP Tools, without advanced security features implemented (feature 1)” is to set up the MCP server, expose to the AI Agent, and have a minimum of 20% of the MCP requirements met. The challenge with this Task is setting up the server, connecting to the agent, and organizing the system. This is the main bulk of this Task. After set up with a hello world tool, more tools will be added to the MCP that cover at least 20% of the FTE requirements.
 - Task 2: The plan for “Implement, test, and demo Prototype AI Agent, with full connection to Prototype Task Execution using Generic MCP Tools (feature 1)” is to set up the AI Agent, LLM backend, and user interface. It should complete at least 3 of the Task-Based Performance Requirements, which require various MCP tools. As such, this task correlates with Task 1 and ensures the AI part of the system is connected and set up. This provides the baseline for future milestones to improve and integrate with the AI.
 - Task 3: For milestone 2, only a proof-of-concept (POC) is required for the Retrieval Augmented Generation. This means, to complete this task for Milestone 2, the AI must be able to ingest large, user-provided documents, and can ingest media such as images. Focus should be on the other Tasks, but to ensure the system baseline is established a POC is required.
 - Task 4: This task ensures that the system begins becoming deployable. So, the system must have a user interface and allow full user interaction with the AI Agent (Task 2) and RAG (Task 3). This user interaction forms the frontend, with the other tasks forming the middle- and backend, ensuring a full system architecture is implemented. Finally, this task requires the code to be pushed to GitHub, allowing tracking and setting the project up for success regarding versatile deployment.
9. Date(s) of meeting(s) with Client during the current milestone
 - 9/18/2026, 9/19/2026
10. Client feedback on the current milestone
 - The AI Agent could act as a virtual crew member in a boat or aircraft.
 - Since the AI Agent has reasoning, it can do analysis. For example, analyzing the current weather for an active mission.

- Really likes the idea.
 - It would be good at Regulations with the RAG capability.
 - Hardware is a concern it will need capable hardware to run a sufficiently intelligent LLM.
11. Date(s) of meeting(s) with Faculty Advisor during the current milestone
- 8/24/2026, 9/18/2026, 9/24/2026
12. Faculty Advisor feedback on each task for the current Milestone
- Task 1: Met with team and is happy with current progress. Believes team is working hard on this task and will be successful.
 - Task 2: Same as above, team has completed this task in full.
 - Task 3: Same as above, team has completed this task in full.
 - Task 4: Same as above, team has completed this task in full.
 - Task 5: Team worked well together on document, including the “owner” and “contributors” format.
 - Task 6: Same as above, team has completed this task well.
 - Task 7: Same as above, team has completed this task well.

Faculty Advisor Signature: _____ Date: _____